CRITICAL THINKING · LEVEL C5 (EVALUATE)

Attention Mechanism & Transformer Discussion Questions

សំណួរពិភាក្សា និងការវាយតម្លៃសុីជម្រៅ

Deep Learning / NLP · CLO2 · Asia Euro University

Master of Science in Information Technology Engineering

CRITICAL THINKING (LEVEL C5)

សំណួរពិភាក្សា និងការវាយតម្លៃសុីជម្រៅ

AEU

១ ហេតុអ្វីត្រូវប្រើ Scaling Factor $\sqrt{d_k}$? តើវាការពារ Vanishing Gradient យ៉ាងដូចម្តេច?

Why do we use the scaling factor $\sqrt{d_k}$? How does it prevent vanishing gradients?

២ បើគ្មាន Positional Encoding តើ Transformer នឹងប្រឈមមុខនឹងបញ្ហាអ្វីខ្លះ?

Without positional encoding, what problems would a Transformer face?

៣ តើ Attention Weights អាចយកមកធ្វើជាភស្តុតាងនៃការរៀបចំវាក្យសម្ព័ន្ធខ្មែរបានដែរឬទេ?

Can attention weights be used as evidence of Khmer syntactic structure?

៤ តើ Attention Weight តែមួយមុខគ្រប់គ្រាន់ក្នុងការបកស្រាយហេតុផលរបស់ Model ទាំងស្រុងដែរឬទេ?

Are attention weights alone enough to fully explain the model's reasoning?

ๅ ហេតុអ្វីត្រូវប្រើ Scaling Factor $\sqrt{d_k}$? តើវាការពារ Vanishing Gradient យ៉ាងដូចម្តេច?

Why use $\sqrt{d_k}$, and how does it prevent vanishing gradients?

$$\text{Attention}(Q,K,V) = \text{softmax}(\, QK^T / \sqrt{d_k}\,)\, V$$

Answer

- If Q and K have mean 0 and variance 1, then **Var(q·k) = d_k**. With d_k = 64, the scores have a standard deviation of about 8.
- Large scores push softmax into **saturation** (almost one-hot), where its gradient **p(1−p) ≈ 0**, so the model stops learning.
- Dividing by $\sqrt{d_k}$ brings the variance back to **1**. Softmax stays in its sensitive range and backpropagation stays **stable**.

Numbers: 3 words, d_k = 64

	Scores	Softmax	Gradient
No scaling	[8, 0, −8]	[.9997, .0003, 0]	≈ **0.0003**
÷ √64	[1, 0, −1]	[.665, .245, .090]	≈ **0.22**



Real-world analogy

A **microphone turned up too loud** clips every sound at maximum, so you lose all detail. $\sqrt{d_k}$ is the volume knob that brings the signal back to a level you can hear.

២ បើគ្មាន Positional Encoding តើ Transformer នឹងប្រឈមមុខនឹងបញ្ហាអ្វីខ្លះ?

Without positional encoding, what problems would a Transformer face?

Answer: the model becomes a "bag of words"

- Self-attention is **permutation-invariant**: shuffle the words and you get the same set of outputs.
- **Can't tell who did what to whom**, so subject and object get confused.
- Negation scope, questions vs statements, and time order all break down.
- Translation and generation produce the **wrong word order**.
- **Worse for Khmer:** Khmer has no conjugation or case endings, so grammatical roles depend almost entirely on **word order (S–V–O)**.

$$\text{Fix: } PE(pos,2i) = \sin(pos/10000^{2i/d}),\ PE(pos,2i+1) = \cos(\ldots)$$

Khmer example: same words, opposite meaning

ស្តេច ទៅ បាញ់ សត្វ

"The king goes to shoot the animal."

សត្វ ទៅ បាញ់ ស្តេច

"The animal goes to shoot the king."

Without PE → both sentences give **identical** representations.

Real-world analogy

Concert seat numbers: the same people in different seats form a different arrangement. PE gives every word its "seat number".

៣ តើ Attention Weights អាចយកមកធ្វើជាភស្តុតាងនៃការរៀបចំវាក្យសម្ព័ន្ធខ្មែរបានដែរឬទេ?

Can attention weights serve as evidence of Khmer syntactic structure?

VERDICT: PARTIALLY. SUPPORTING EVIDENCE, NOT PROOF

✓ What supports it

- Some BERT/XLM-R heads line up with **subject → verb** and **adjective → noun** links (Clark et al., 2019).
- Middle layers tend to encode syntax, and deeper layers encode meaning.
- BertViz makes these patterns visible.

✗ Limits for Khmer

- Khmer has **no spaces between words**, so subword tokens split words and attention has to be merged back.
- Many heads send most of their attention to **[CLS] / [SEP] / punctuation**.
- Patterns differ between heads and layers.
- Shows **correlation, not causation**.

How to make it scientific

- Compare with a **Khmer dependency treebank**.
- **Probing classifiers** on the hidden states.
- **Head ablation**: switch a head off and see whether the syntax task breaks.
- Test on many sentences, not just one example.

Example

In «ស្តេចទៅបាញ់សត្វ», a head where បាញ់ (shoot) attends strongly to ស្តេច (king) *suggests* a subject–verb link. But if ស្តេច is split into subword tokens by XLM-R, or another head shows the opposite pattern, the conclusion becomes weak.

៤ តើ Attention Weight តែមួយមុខគ្រប់គ្រាន់ក្នុងការបកស្រាយហេតុផលរបស់ Model ទាំងស្រុងដែរឬទេ?

Are attention weights alone enough to fully explain the model's reasoning?

VERDICT: NO. ATTENTION SHOWS WHERE THE MODEL LOOKED, NOT WHY IT DECIDED

Why not

- **Residual connections & feed-forward layers** carry a lot of information that attention maps don't show.
- **12 layers × 12 heads = 144 maps** in BERT-base, all mixed together. One map is a tiny piece of the picture.
- Very different attention distributions can give the **same prediction** (Jain & Wallace, 2019, "Attention is not Explanation").
- Weights ignore the **size of the value vectors**, so high attention doesn't always mean high impact.

Better explanation tools

Method	Idea
Attention rollout / flow	Combine attention across all layers
Integrated Gradients	Gradient-based importance of each token
SHAP / LIME	Remove words and measure the change
Probing & ablation	Test what is encoded and what is causal

Real-world analogy

Eye-tracking a student shows they looked at the right line of the exam, but it doesn't prove they understood it. Attention is the model's "eye-tracking".

SUMMARY

Key takeaways for the C5 discussion

① $\sqrt{d_k}$ = variance control

It keeps softmax out of saturation, so the gradients stay healthy (≈0.22 instead of ≈0.0003) and training stays stable.

② Positional encoding = word order

Without it, a Transformer is a bag of words: «ស្តេចទៅបាញ់សត្វ» = «សត្វទៅបាញ់ស្តេច». This is critical for order-based languages like Khmer.

③ Attention ≈ a clue about Khmer syntax

It can support a claim about syntax. Tokenization, special tokens and head variation limit it, so validate with treebanks, probing and ablation.

④ Attention ≠ full explanation

Correlation, not causation. Combine it with rollout, gradients, SHAP and ablation to explain the model's reasoning.

C5 mindset: don't just *describe* the attention map. *Evaluate* what it proves, what it doesn't, and what extra evidence you need.